

ECG TRAINING

- ECG MACHINES (3.6 AND 12 CHANNELS)
- CALIBRATIONS AND ECG GRAPH PAPER
- REASONS TO ALTER AMPLITUDE
- 20 ELECTRODES AND 12 LEAD ECG
- PLACEMENT OF ELECTRODES



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OBJECTIVES

AT THE END OF THE TRAINING ,PARTICIPANTS SHOULD BE ABLE TO:

- IDENTIFY THE MACHINES BY THEIR PAPER WIDTH OR SIZE
- SHOULD KNOW THE STANDARD CALIBRATION OF THE ECG MACHINES
- SHOULD KNOW WHEN TO ALTER THE CALIBRATION WHILE RECORDING.
- SHOULD BE ABLE TO PLACE THE 10 ELECTRODES FOR 12 LEAD RECORDING CORRECTLY.

PRE-TRAINING QUESTION

1. WHAT TYPE OF MACHINE PRODUCES THE RECORDING HERE

A. 3 CHANNELS B. 6 CHANNELS. C. 12 CHANNELS

2. AT WHAT AMPLITUDE IS THE RECORDING DONE.

A. 5mm/mv B. 15mm/mv. C. 10mm/mv

3. WHAT IS THE SPEED OF THE RECORDING?

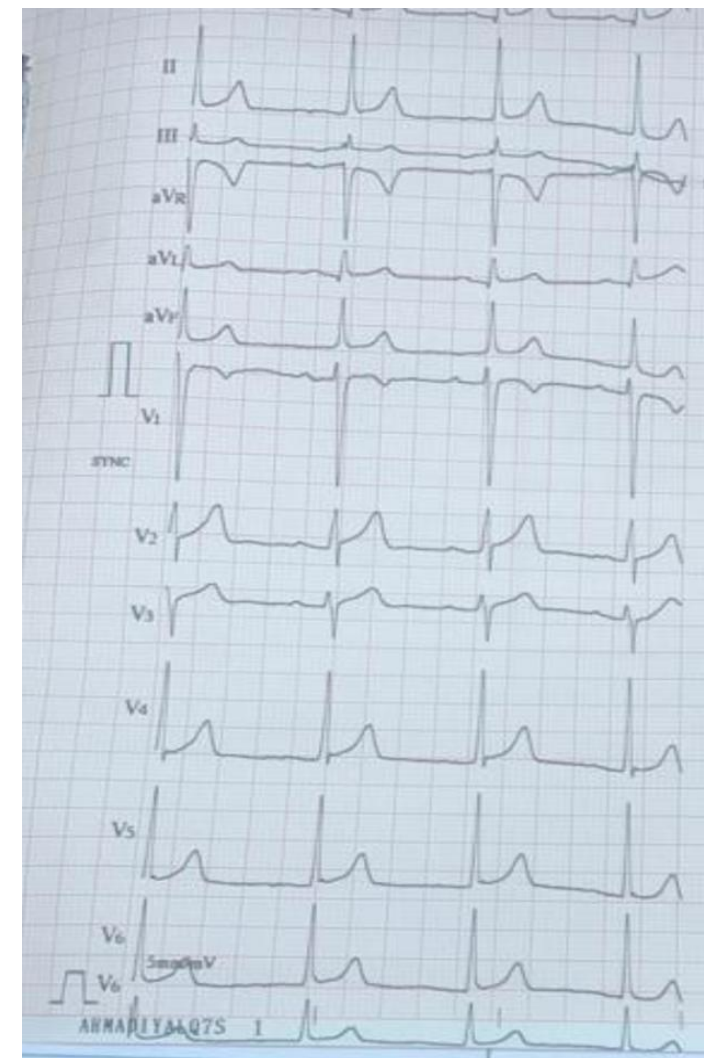
A. 25MM/S. B. 25M/S. C. 25SEC

4. HOW MANY LEAD RECORDINGS DO WE HAVE ?

A. 10. B. 12. C. 14

5. LEAD II RECORDS BETWEEN

A. RA&LA . B. RA&LA. C. RA&LL



BRIEF HISTORY OF ECG

EARLY FOUNDATION WAS IN THE 1800s.

LUIGI GALVANI (1791) FIRST DISCOVERED THAT MUSCLES PRODUCE ELECTRICITY.

CARLO MATTEUCCI (1840s) DEMONSTRATED THAT ELECTRICAL CURRENTS ACCOMPANY EVERY HEARTBEAT.

INVENTION OF ECG WAS IN THE EARLY 1900s.

WILLEN EINTHOVEN (1903) IS A DUTCH PHYSIOLOGIST WHO INVENTED THE FIRST PRACTICAL ELECTROCARDIOGRAPH

HE DEVELOPED THE STRING GALVANOMETER A SENSITIVE DEVICE THAT COULD RECORD ELECTRICAL ACTIVITY FROM THE HEART

HE CREATED THE STANDARD LEAD SYSTEM(I,II AND III) STILL IN USE TODAY.

WON THE NOBEL PRIZE IN MEDICINE IN 1924.



CARLO MATTEUCCI



LUIGI GALVANI



WILLEN EINTHOVEN



GALVANOMETER

The old machine was called string galvanometer

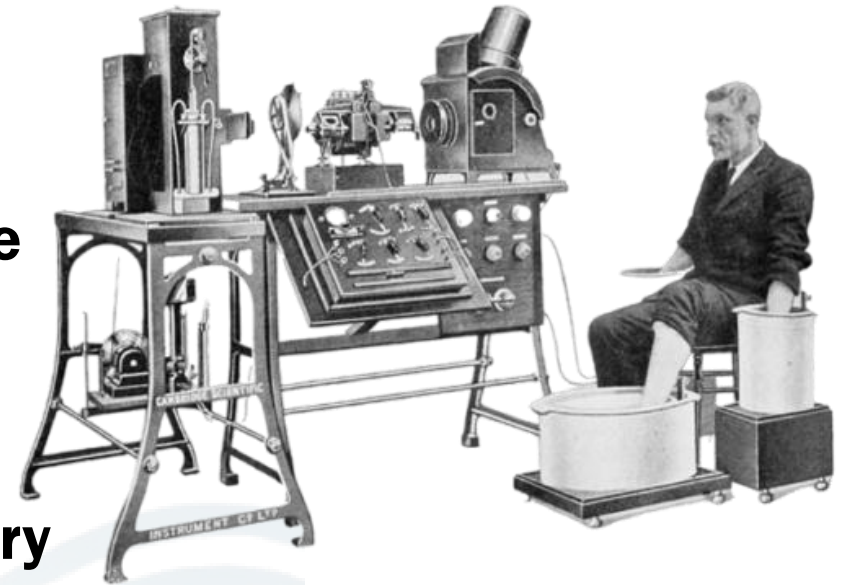
It weighed over 270kg

Required five operators to run

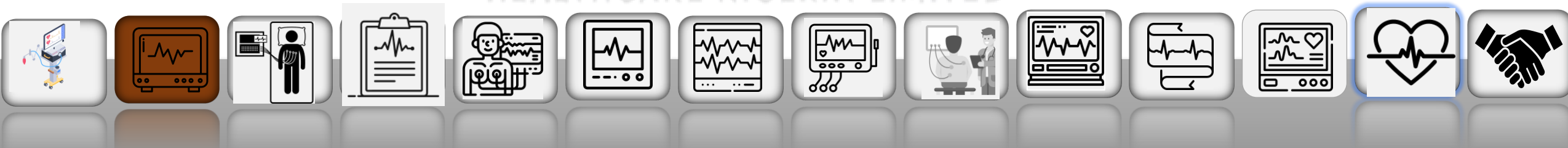
Patients had to immerse their hands and legs in saline bath to record ECG.

By the 1920s-1930s, ECG machine became more portable and user friendly

Modern ECG machines are now digital ,portable ,battery powered,12 lead display screen and automatic interpretation display features –not really reliable totally.



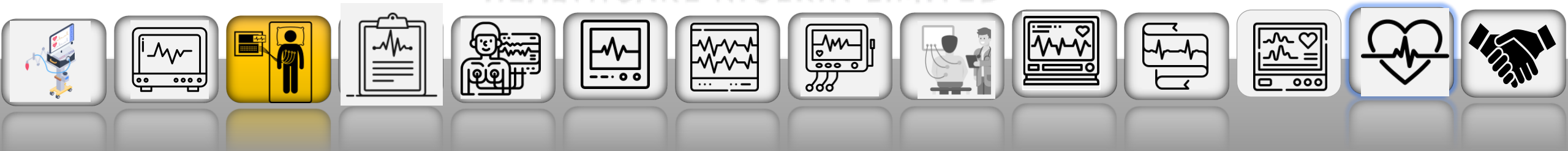
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THE ECG MACHINES

The ECG machine is a diagnostic tool used to measure the electrical activities of the heart.

-We have the 3,6 and 12 channels machines



THE 3-CHANNELS MACHINES

RECORDS AND PRINTS 3 LEADS OF ECG DATA AT A TIME FROM THE STANDARD 12 LEADS.

CYCLES THROUGH THE 12 LEADS IN BATCHES

IT DISPLAYS 3 LEADS AT A TIME EXCLUDING THE RHYTHM LEAD.

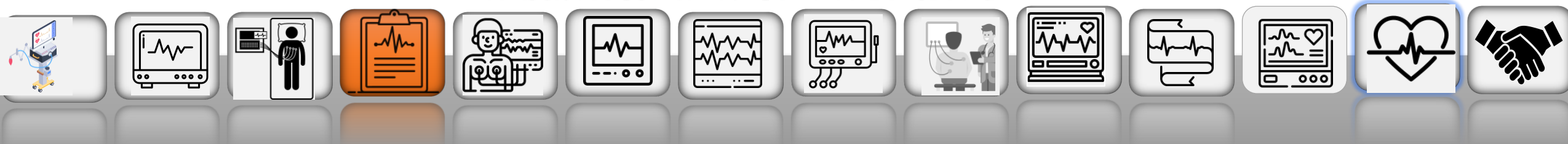
THEY TYPICALLY USES 63MM OR 80MM PAPER WIDTH

MOSTLY USED IN AMBULANCES AND FIELD CLINICS BECAUSE THEY ARE PORTABLE AND VERY HANDY.

THE PAPERS ARE MOSTLY THE ROLLING TYPE OF PAPER WHILE FEW MACHINES USES THE FLAT SQUARE PAPERS EG SCHILLER 3 CHANNELS.



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THE 6-CHANNELS MACHINES

UNLIKE THE 3 CHANNELS, THE 6 CHANNELS PRINTS AND RECORDS 6 LEADS OF ECG WAVEFORMS SIMULTANEOUSLY.

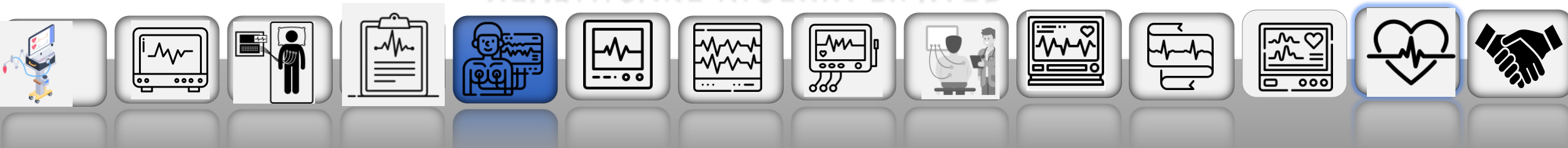
IT IS FASTER AND MORE DETAILED THAN 3 CHANNELS AND MORE AFFORDABLE THAN THE 12 CHANNELS

PRINTS THE 12 LEADS IN TWO BATCHES.

USES 110MM OR 210MM WIDE THERMAL PAPER (ROLL OR FLAT SQUARED SHAPED Z-FOLD)



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THE 12 CHANNELS MACHINE

RECORDS SIGNALS FROM 12 LEADS IN THE FORMAT LIKE 3X4,6X2 OR 12X1

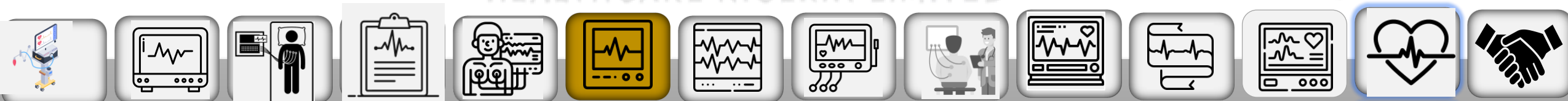
PROVIDES A COMPREHENSIVE VIEW OF THE HEARTS ELECTRICAL ACTIVITIES

ALL LEADS ARE RECORDED AT THE SAME TIME OFFERING A SNAPSHOT OF HEART FUNCTION.

WIDTH OF THE PAPER IS 210MM(A4 WIDTH)

LENGTH VARIES DEPENDING ON MACHINES AND PRINT FORMAT(COMMONLY 140MM,295MM OR CONTINUOUS ROLL),THE WIDTH FOR THE ROLL PAPER IS 80 OR 112MM

PAPER CAN BE Z-FOLD OR FANFOLD OR THE ROLLING PAPER



CALIBRATIONS IN ECG

CALIBRATION IN ECG REFERS TO THE STANDARD SETTINGS USED TO ENSURE THAT THE ECG TRACING ACCURATELY REPRESENTS THE HEARTS ELECTRICAL ACTIVITY IN BOTH AMPLITUDE(VOLTAGE) AND TIME(DURATION).

PROPER CALIBRATION IS ESSENTIAL FOR INTERPRETING ECGs CORRECTLY

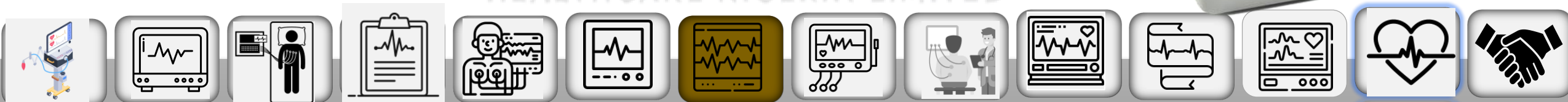
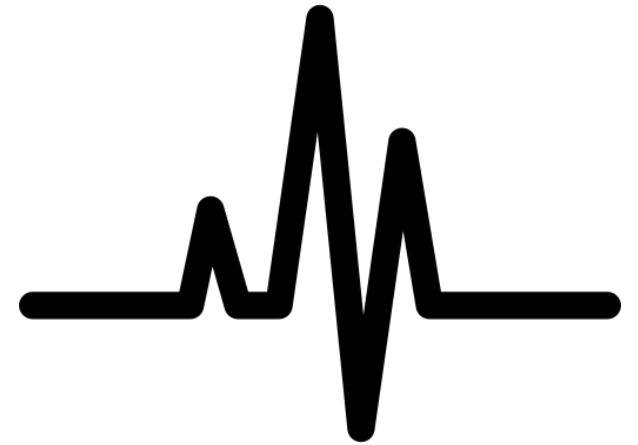
STANDARD GAIN OR AMPLITUDE IS 10MM/MV. THAT MEANS THAT A 1MILLIVOLT(MV) ELECTRICAL SIGNAL WILL PRODUCE A 10MILLIMETER(MM) DEFLECTION ON ECG PAPER OR SCREEN.

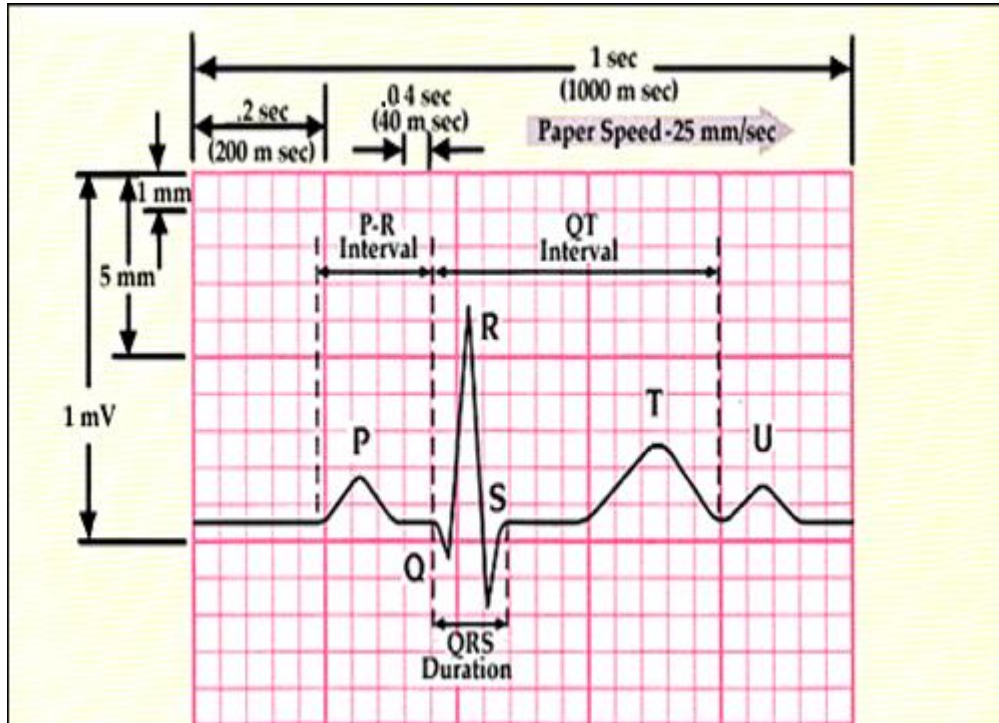
ON ECG PAPER,1 SMALL BOX=1MM;1MV=10 SMALL BOXES OR 2 LARGE BOXES

TIME WHICH IS THE PAPER SPEED IS 25MM/S

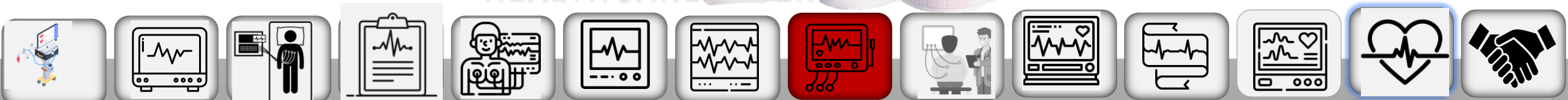
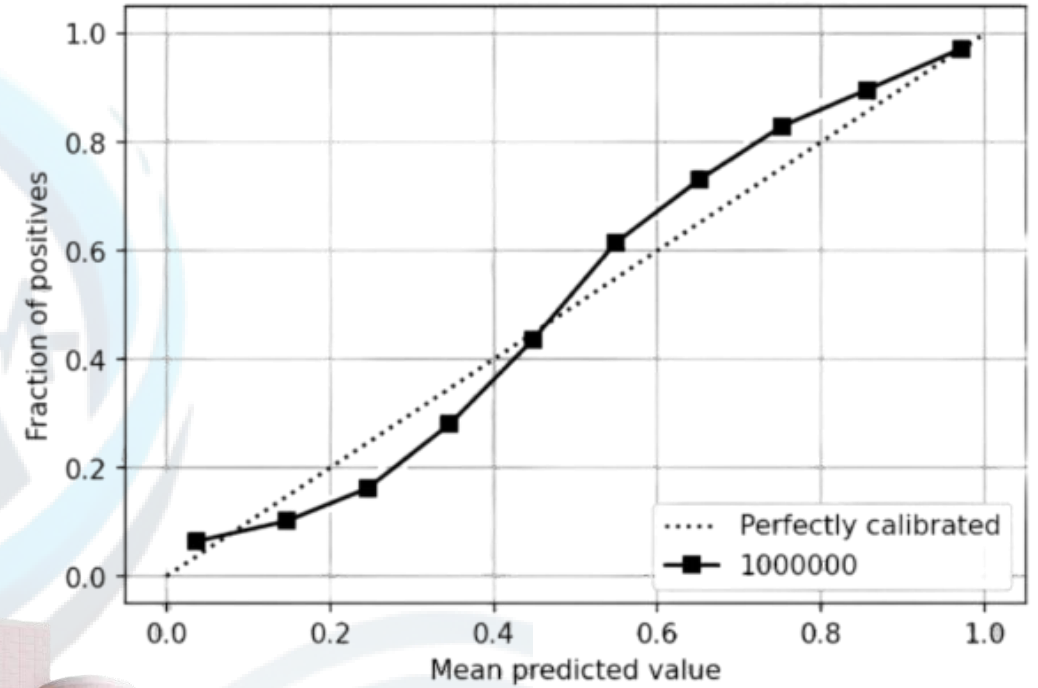
EACH SMALL BOX=0.04s(40ms)

EACH LARGE BOX(5 SMALL BOXES) 0.20s(200ms)





Calibration plots (Reliability Curves)



CALIBRATION-REASONS

VALIDATION BEFORE INTERPRETATION-ENSURES MEASUREMENTS OF WAVES AMPLITUDE AND INTERVALS ARE ACCURATE.

DETECTING ADJUSTMENTS-IF ECG TRACING LOOKS ODDLY LARGE/SMALL OR STRETCHED,CHECK THE MARK-IT MAY INDICATE NON-STANDARD SETTING(LIKE HALF VOLTAGE OR DOUBLE SPEED)

TO CHECK CALIBRATION,LOOK AT THE BEGINNING/ END OF EACH LEAD STRIP FOR THE VERTICAL CALIBRATION SPIKES

IF IT IS APPROXIMATELY 2 LARGE BOXES(10MM)HIGH AND 1 BOX WIDTH,IT'S STANDARD.

SMALLER,BIGGER OR WIDE MARKS INDICATES ALTERED CALIBRATION.

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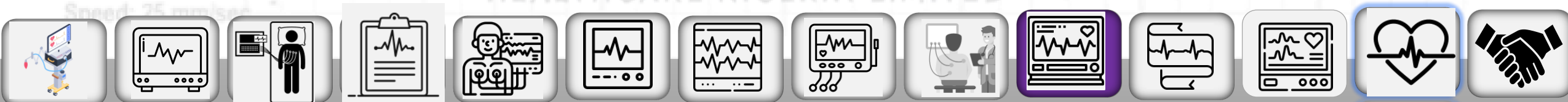


ALTERNATE CALIBRATION SETTINGS

**CALIBRATION SETTINGS ARE ALTERED WHEN;
LARGE QRS COMPLEXES OVERLAPPING EACH
OTHER-ALTER TO 5MM/MV OR 2.5MM/MV SO
THE SIZE OF THE WAVE IS REDUCED HERE.**

**SMALL VOLTAGES, EG IN OBESITY, PERICARDIA
EFFUSION, YOU MAY WANT TO INCREASE
AMPLITUDE TO 20MM/MV**

**IN ALL OR MOST CASES, THE SPEED WHICH IS
25MM/S IS NEVER ALTERED.**



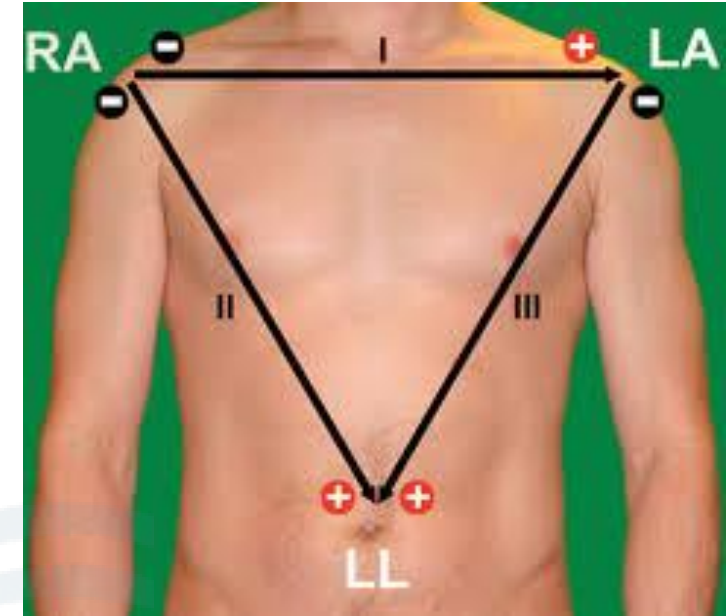
10 ELECTRODES 12 LEADS ECG

10 ELECTRODES ARE THE SUCTIONS OR DISPOSABLE ELECTRODES AND THE CLAMPS PLACED ON THE BODY TO MAKE THE 12 LEAD RECORDINGS.

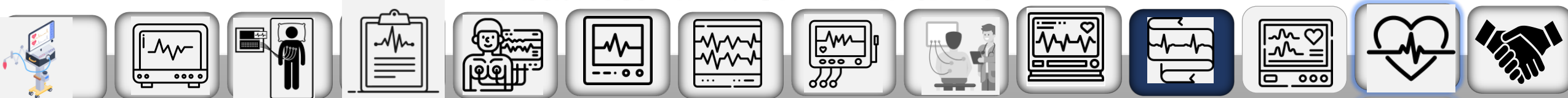
WE HAVE 6 CHEST ELECTRODES AND 4 LIMB CLAMPS OR ELECTRODES

THE CHEST ELECTRODES ARE LABELLED V1-V6 AND HAVE SPECIFIC POSITIONS ON THE INTERCOSTAL SPACES WHERE THEY ARE PLACED.

THE LIMB ELECTRODES ARE 4 AND ARE COLOUR CODED(RED,YELLOW,GREEN AND BLACK) OR WITH THEIR OWN LABELS TOO(R OR RA-RIGHT ARM,L OR LA-LEFT ARM,F OR LF-LEFT FOOT AND N OR RL-NEUTRAL FOR RIGHT FOOT)



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THE 12 LEADS ARE THE RECORDINGS FROM THE 10 ELECTRODES PLACED ON THE SURFACE OF THE BODY.

OFTEN TIME WE MISTAKE THE ELECTRODES FROM THE LEADS.

10 ELECTRODES PRODUCES 12 LEADS RECORDINGS

THE MONITORS IN THE ICU ETC DOES NOT PRODUCE 12 LEADS BUT RATHER IN MOST CASES RHYTHM STRIP BUT FOR ACCURATE DIAGNOSIS IN MOST CASES, THE 12 LEADS ARE NEEDED.

THE 12 LEADS ARE –V1-V6,I,II,III,aVR,aVF,AvL.

V1-V6 REPRESENTS THE CHEST LEADS

I,II AND III ARE BIPOLAR LEADS

aVR,aVF and aVL, ARE AUGMENTED LEADS

Limb Leads (6 leads)

- **Standard Bipolar Leads (3):**

1. $\text{Lead I} = \text{LA} - \text{RA}$

2. $\text{Lead II} = \text{LL} - \text{RA}$

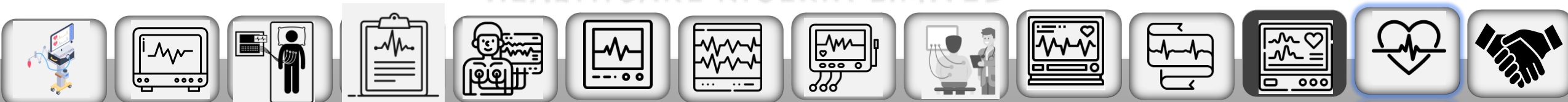
3. $\text{Lead III} = \text{LL} - \text{LA}$

- **Augmented Unipolar Leads (3):**

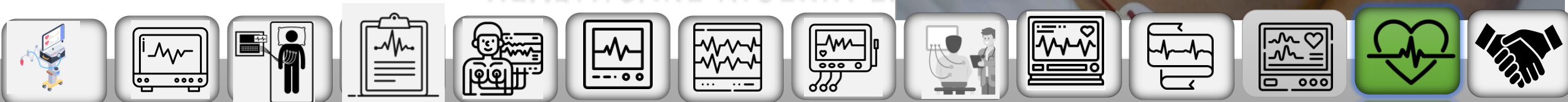
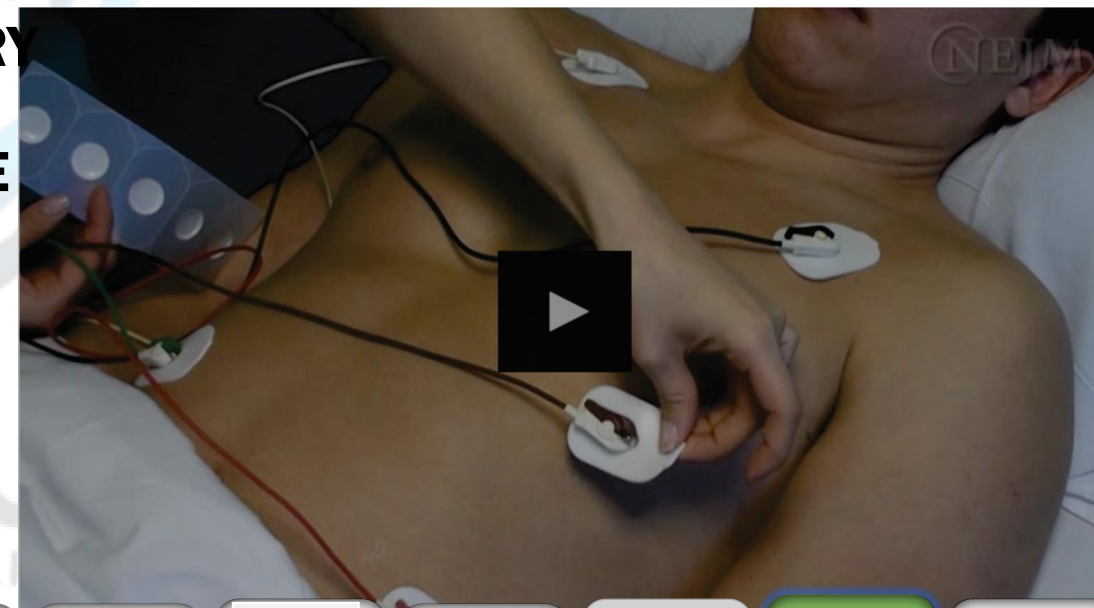
4. $\text{aVR} = \text{RA} - (\text{LA} + \text{LL})/2$

5. $\text{aVL} = \text{LA} - (\text{RA} + \text{LL})/2$

6. $\text{aVF} = \text{LL} - (\text{RA} + \text{LA})/2$



RL OR N OR BLACK-RIGHT FOOT.



APPRECIATION

**“ NO MAN IS AN EMBODIMENT OF
KNOWLEDGE AND WISDOM”**



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